Structured Validation for Life Sciences AI

A Collaboration Framework

Sage BioNetworks Follow-Up

Don B. | February 2026

What We Heard

No Baseline Exists

There is no standardized process for comparing LLM tool outputs in life sciences. Teams evaluate tools ad hoc, making procurement and adoption decisions on anecdote rather than evidence.

Use Cases Must Be Persona-Driven

Different researchers need different things. A trainee learning pathway biology has fundamentally different needs than a PI orchestrating a multi-site collaboration.

Structured Testing Matters

Sage teams want reproducible test cases with known answers — not cherry-picked demos. Gold-standard questions with verifiable paths are the currency of trust.

Language Needs Translation

Technical capabilities must be expressed in terms researchers actually use: "Can it find drugs for my target?" not "Does it resolve HGNC IDs?"

The Baseline Challenge

Current State

- Ad hoc tool evaluation
- Cherry-picked demos
- No reproducible test cases
- Subjective quality assessment
- No cross-tool comparison framework

What a Baseline Requires

- Standardized questions with known answers
- Entity resolution to canonical IDs (CURIEs)
- Multi-dimensional quality scoring
- Evidence grading with confidence levels
- Reproducible across tools, models, and time

The gap between these columns is exactly what our platform addresses.

Research Evaluation Benchmarks

Competency Questions as Standardized Test Cases

Benchmark	Domain	Key Entities (CURIEs)	Complexity
FOP Drug Mechanism	Drug Mechanisms	Palovarotene (CHEMBL:2105648) RARG (HGNC:9866), ACVR1 (HGNC:171)	Low 4-node linear path
ARID1A Synthetic Lethality	Precision Oncology	ARID1A (HGNC:11110) EZH2 (HGNC:3527), Tazemetostat (CHEMBL:3414621)	Medium Protein complex + drug + trial
TP53 Synthetic Lethal Pairs	Scalable Validation	TP53 (HGNC:11998) TYMS (HGNC:12441), 5-FU (CHEMBL:185)	High 209 SL pairs, CRISPR validation
Health Emergency Landscape	Clinical Landscape	18,636+ recruiting trials across oncology, diabetes, Alzheimer's, Long COVID	Medium Real-time trend detection

15 benchmarks spanning 8 therapeutic domains — each with gold-standard paths and verifiable answers

10-Dimension Quality Framework

#	Dimension	What It Measures	Range
1	**CURIE Resolution**	Entities mapped to canonical identifiers (HGNC, CHEMBL, MONDO)	0-10
2	**Source Attribution**	Every claim traceable to a named API + identifier	0-10
3	**Evidence Grading**	Claims tagged L1-L4 with confidence ranges	0-10
4	**Hallucination Risk**	Zero fabricated entities, relationships, or trial IDs	0-10
5	**Multi-Source Triangulation**	Findings confirmed across 2+ independent databases	0-10
6	**Clinical Relevance**	Results connect to actionable drugs or active trials	0-10
7	**Pathway Completeness**	Full mechanistic paths discovered (drug > target > disease)	0-10
8	**Contradiction Handling**	Conflicting sources identified and resolved transparently	0-10
9	**Temporal Currency**	Data reflects current trial status and approval landscape	0-10
10	**Knowledge Graph Density**	Captured interaction breadth (nodes, edges, cross-refs)	0-10

Pass threshold: 35/40 on dimensions 1-4 (non-negotiable) + 30/60 on dimensions 5-10

The Fuzzy-to-Fact Protocol

7-Phase Structured Workflow



HGNC · ChEMBL · UniProt · Ensembl · STRING · BioGRID · Open Targets · PubChem · WikiPathways · ClinicalTrials.gov · PubMed · Entrez · IUPHAR

13 biomedical APIs orchestrated across 7 phases with dedicated specialist agents

Production Results

5	9.3/10	9.7/10	0	100%
Reports Generated	Median Quality	Protocol Execution	Hallucinations	NCT Verification

Across 5 production reports covering drug mechanisms, synthetic lethality, gene networks, doxorubicin toxicity, and clinical landscape analysis

Case Study: Doxorubicin Cardiotoxicity in NF1

Directly relevant to Sage BioNetworks' Neurofibromatosis research

Knowledge Graph Summary

- **18 nodes** (11 genes, 3 compounds, 3 pathways, 2 diseases)
- **20 edges** (verified relationships)

3 Cardioprotective Axes:

- **TOP2B** degradation via Dexrazoxane (L4)
- **NRF2/KEAP1** antioxidant response (L2)
- **MEK inhibition** via Selumetinib (L4)

Synapse Grounding

- **8 Synapse datasets** identified
- **56% node coverage** (10/18 grounded)
- **35% edge coverage** (7/20 grounded)

Key Datasets:

- **syn200548**: Rat heart doxorubicin/dexrazoxane
- **syn44222814**: NF1 Schwann cell + selumetinib
- **syn61259950**: Phase II Selumetinib (NCT02407405)

Three Research Personas

Educational / Trainee

"I'm learning pathway biology and need to understand how drugs connect to diseases"

Benchmark:
FOP Drug Mechanism — 4-node linear path

Value:
Structured learning with verified relationships

"Show me the mechanism step by step"

Independent Investigator

"I need to validate my hypothesis about synthetic lethal targets before writing a grant"

Benchmark:
ARID1A Synthetic Lethality — multi-source validation

Value:
Multi-database triangulation with evidence grading

"Cross-reference my targets against all data"

Project Manager

"I need to understand the clinical landscape to prioritize resources"

Benchmark:
Health Emergency Landscape — 18,636+ trials

Value:
Rapid clinical landscape with structured output

"Give me an evidence-based overview for leadership"

Three Value Propositions for Sage

Structured Validation & Testing

- 15 benchmark questions with gold-standard answers
- 10-dimension quality scoring framework
- Reproducible test suite for any LLM tool comparison

The baseline Sage has been looking for

Data Context & Curation

- Automated Synapse dataset grounding
- Knowledge graph overlay on existing Sage data
- Evidence-graded relationships with CURIE provenance

Turn Synapse data into structured knowledge

Unifying the Research Pipeline

- Fuzzy question to validated knowledge graph
- 13 APIs in reproducible 7-phase protocol
- Persona-driven entry points for researchers

One protocol, from bench question to clinical context

Evidence Grading: L1-L4 Pyramid



Synapse Grounding Upgrade

When a KG node or edge is grounded in a Synapse dataset, its evidence level can be upgraded.

Example:
NRF2/KEAP1 at L2 → grounded by syn205590 (Nrf2 KO mouse) → upgradeable to L3

Proposed Collaboration: 3 Phases, 12 Weeks

Phase 1: Proof of Concept

Weeks 1-4

- Run 3-5 NF1-relevant benchmarks end-to-end
- Evaluate against 10-dimension quality framework
- Validate Synapse grounding on NF1 datasets
- Deliverable: Quality report + knowledge graphs

Phase 2: Benchmark Expansion

Weeks 5-8

- Co-develop 5-10 Sage-specific competency questions
- Add Synapse-native data sources to API orchestration
- Train evidence grading on Sage domain expertise
- Deliverable: Expanded catalog + Synapse integration spec

Phase 3: Integration Pilot

Weeks 9-12

- Deploy platform for 3-5 Sage researchers
- Collect structured feedback on usability and accuracy
- Measure: time-to-insight, evidence quality, coverage
- Deliverable: Pilot report + roadmap for 2026 H2

Proposed Next Meeting Agenda

Section	Topic	Time	Lead
1	Review this deck + Q&A on platform capabilities	20 min	Don B.
2	Sage priorities: Which NF1 questions matter most?	20 min	Sage team
3	Live demo: Run one Sage-chosen question through the protocol	25 min	Don B.
4	Discuss collaboration framework + next steps	15 min	Joint
	Total	**80 min**	

Flexible format — happy to adjust based on Sage's priorities and available time

Thank You

Let's build the baseline for life sciences AI evaluation

Benchmark Catalog
15 questions across 8 domains

Quality Framework
10-dimension L1-L4 grading

NF1 Case Study
18 nodes, 20 edges, 8 Synapse datasets

Built on open standards: BioLink model, CURIEs, ClinicalTrials.gov v2, Synapse.org

Don B. | LangGraph + 13 Biomedical APIs + Synapse Integration